

Safety Rules For Filing With Different Files



1. Never use file without handle securely attached.
2. Apply pressure only on forward stroke.
3. Place one hand on the handle and the other on file tip.
4. Do not file too rapidly.
5. Clean clogged file card or wire brush.
6. Never hammer or pry with file.
7. Never carry files in your pockets.
8. Keep your files in good condition.
9. Use the right file for the job.
10. Use a file with a rough or coarse cut to quickly remove materials.

Video :

- <https://youtu.be/lOR2UwZRBws>

Feedback : Activity

Kwa Thema Skills School

Subject : Maintenance

Marks : 10

Grade : Year 3

Time : 30 mins

Date : / /

Question : 1

1. Complete the following basic safety rules when using different files.
 - 1.1 Never use file without handle _____ attached.
 - 1.2 Apply _____ only on forward stroke.
 - 1.3 Place one _____ on the handle and the other on file tip.
 - 1.4 Do not _____ too rapidly.
 - 1.5 Clean clogged file card or _____ brush.
 - 1.6 Never _____ or pry with file.
 - 1.7 Never carry files in your _____.
 - 1.8 Keep your files in _____ condition.
 - 1.9. Use the _____ file for the job.
 - 1.10 Use a file with a _____ or coarse cut to quickly remove materials.

[file: pressure: pockets: right: rough: hand; wire: good: securely: hammer]

Common Metalwork Equipments or Tools.

1. Centre Punch



2. Hammer



3. Pliers



4. Bench Vice



5. Tinsnip



6. Steel Rule



Kwa Thema Skills School

Subject : Maintenance

Marks : 6

Grade : Year 3

Time : 30 mins

Date : / /

Question:1

1. Identify the following common equipments in Metalwork workshop.



1.1



1.3



1.5



1.2



1.4



1.6

(6)

Prevent Metal From Rust Using Relevant materials .

1. Use an Alloy



2. Apply Oil



3. Store Properly



4. Apply Dry Coating



5. Paint the Material



6. Galvinize



7. Powder Coating



8. Careful Handling To Scratching



9. Blueing



10. VCI Packaging (Vapour Corrosion Inhibitors)



Procedure of Rust Prevention Permanently

Step 1 : Check structural damage to the Metal.



Step 2 : Scrape off excess rust.



Step 3 : Apply the relevant rust prevention material.



Feedback : Activity

Kwa Thema Skills School

Subject : Maintenance

Grade : Year 3

Date : / /

Marks : 10

Time : 30 mins

Question:1

1. Identify the following methods of rust prevention.



1.1 _____



1.2 _____



1.3 _____



1.4 _____



1.5 _____



1.2 _____



1.3 _____



1.4 _____



1.9 _____



1.10 _____

Kwa Thema Skills School

Subject : Maintenance

Grade : Year 3

Date : / /

Marks : 10

Time : 30 mins

Question:1

1. Identify the following engine parts or components.



1.1 _____



1.2 _____



1.3 _____



1.4 _____



1.5 _____



1.2 _____



1.3 _____



1.4 _____



1.9 _____



1.10 _____

Cutting Sheet Metal Using Appropriate Tools-Tinsnip.

1. Multi-Purpose Tin Snip



- It is used for cutting softer materials.

3. Duckbill Pattern Tin Snip



- It is used to cut sharper curves and circles.

5. Spring Loaded Tin Snip



- It is used to give maximum force exertion.

2. Straight Cut Tin Snip



- It is used for cutting curves and irregular shapes.

4. Bulldog Pattern Tin Snip



- It is used for cutting and trimming.

6. Carved Tin Snip



- It is used for cutting arch, bends, circle.

7. Jewellers Tin Snip



- It is used for cutting lightweight sheet metals.

9. Utility and Multi-purpose Tin Snip



- It is used for cutting and trimming workshop items

Video :

- <https://youtu.be/mV1QntAvAbo>

Feedback : 1. Activity

2. Demonstration

8. Universal Tin Snip



- It is used for cutting straight and curved lines.

10. Straight Pattern Tin Snip



- It is used for application of straight cut..

Kwa Thema Skills School

Subject : Maintenance

Grade : Year 3

Date : / /

Marks : 10

Time : 30 mins

Question:1

1. Give the purpose of the following Tinsnip

Multi-Purpose Tinsnip



1. _____

Duckbill Pattern Tinsnip



3. _____

Straight Cut Tinsnip



2. _____

Bulldog Pattern Tinsnip



4. _____

Spring Loaded Tinsnip



Carved Tinsnip



Jewellers Tinsnip



Universal Tinsnip



Utility and Multi-Purpose Tinsnip



Straight Pattern Tinsnip



How to Light And Adjust A Gas Torch



1. Give cylinder a once-over.
2. Check for torch damage.
3. Install reverse flow check valve or flash back arrestors.
4. Confirm the work area well ventilated.
5. Slowly open the oxygen cylinder to blow out any debris.
6. Attach and tighten the regulators to the cylinders.
7. Install hoses.
8. Pressurize the system.
9. Leak test all connections.
10. Adjust the regulators for the tip being used.

Video :

- <https://youtu.be/9zYMV4s1E0E>

Feedback : 1. Activity

2. Demonstration

Kwa Thema Skills School

Subject : Maintenance

Grade : Year 3

Date : / /

Marks : 10

Time : 30 mins

Question : 1

1. Fill in the missing words.

1. Give _____ a once-over.
2. Check for _____ damage.
3. Install _____ flow check valve or flash back arrestors.
4. Confirm the work area well _____.
5. Slowly open the _____ cylinder to blow out any debris.
6. Attach and tighten _____ to the cylinders.
7. Install _____.
8. _____ the system.
9. Leak test all _____.
10. _____ the regulators for the tip being used.

[Adjust: pressurize: connections: cylinder: reverse: torch: hoses: ventilated: oxygen: regulators]

(10)

Solder Any Material

Step 1 : Make sure the torch is in a well well ventilated area.



Step 3: Hold the striker in one hand and the torch in the other.



Step 5 : Adjust the torch to obtain the the correct flame.



Step 2 : Connect propane tank to the torch
And open the valve.



Step 4 : Hold and use the striker.



Step 6 : Begin the soldering process.



Step 7 : Safe practices when soldering project is complete.



Video :

- https://youtu.be/XzdQk_SAUoQ

Feedback : 1. Activity

2. Demonstration

Kwa Thema Skills School

Subject : Maintenance

Grade : Year 3

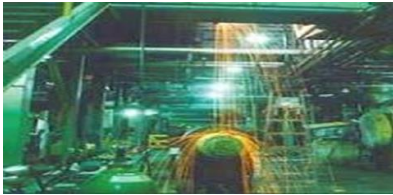
Date : / /

Marks : 10

Time : 30 mins

Question:1

1. Steps of Soldering any material.



Stage 1 : _____



Stage 3: _____



Stage 5 : _____



Step 2 : _____



Step 4 : _____



Step 6 : _____



Stage 7 : _____

2. Name three equipments that are used in soldering.

2.1 _____

2.2 _____

2.3 _____

(7)

(3)

[10]